National University of Computer and Emerging Sciences, Lahore Campus

Course:	Programming Fundamentals	Course Code:	CS 118
Program:	BS (CS, SE)	Semester:	Spring 2021
Duration:	180 Min	Total Marks:	100
Paper Date:	14-Jul-2021	Page(s):	18
Section:	ALL	Section:	
Exam:	Final	Roll No:	

Roll Number: _____

Instructions:
Attempt all questions
You might use extra sheets for working but do not attach them.
Write the final answer in the space provided for it.
Questions are NOT ALLOWED during exam, take reasonable assumptions where needed.

Question No.	Marks	Marks Obtained
Q1	10	
Q2	35	
Q3	10	
Q4	10	
Q5	35	
Total	100	

Q No. 1: Write the output of the following program segment. Identify errors (if any). Assume all the code snippet contains #include<iostream>, using namespace std; and main, and **overwriting / cutting** is not allowed.
[10 Marks]

10

a)

```
int x=10;
while(x++ < x+1)
cout<<"Pakistan"<<x;
```

Output:
Infinite Loop.

b)

```
float x = -10;
while (x){
    ++x;
    int x = 20;
    x *= 10;
    x += 10;
}
cout << x << endl;
```

Output:
0

c)

```
int i, j, sum = 10;
  for (i = 0; i<5; i++)
    if (i % 2)
        for (j = 0; j <= 3; sum += j, j++);
    else
        for (j = 3; j>0; sum += j, j--);
cout << sum;
```

Output:
40

d)

```
int main()
{
   char a[20]=”welcome to punjab”;
   a[10]=’\0’;
   cout<<a;
}
```

Output:
welcome to

e) Define Conditional Operator and its syntax.

Condition operator (?) is ternary operator and its syntax is;
(Condition)? A:B;

Q No. 2: Please note there are no syntax errors in the given set of codes. All the code snippet contains #include<iostream> and using namespace std; and **overwriting/ cutting** is not allowed.

35

1. What would be the output produced by executing the following C++ code? **[2 Mark]**	

int main()
{
int a=5;
char b=a;
cout << a/b;
}
	Output:
	1

2. What would be the output produced by executing the following C++ code? **[2 Mark]**	

int main()
{
int a = 0, b=36;
float f=3.9;
b+=(a = 50)*f/3+5.1-6.8*10-b%5;
cout << a << "$" << b;
return 0;
}
	Output:
	50$37

3. What would be the output produced by executing the following C++ code? **[2 Mark]**

```
main( ) {
int x = 1, y =2, n=50 ;
if ( n%y && n%3 && n%4 || n%5  )
    { n=n/y;
      cout<<"Hello World \n" ;}
cout<<n <<": "<< n%3 <<": "<< n%5 ;
return 0;
}
```

Output:

```
50: 2: 0
```

4. What would be the output produced by executing the following C++ code? **[3 Mark]**

```
int counter (int value)
{
 static int count =0;
 count = count +value;;
 return count;
 }
int main()
{
  int i , j;
  for (i=0; i <=5; i++)
     j = counter(i);
   cout<<"J = " <<j<<endl;
 return 0;
}
```

Output:

```
J = 15
```

5. What would be the output produced by executing the following C++ code? **[3 Mark]**

```
main( )
{
int x, y;
x = 1;
y = ++x * ++x;
cout << x <<" :"<< y <<endl;
x += 5;
y = (x+=1) * (x+1);
cout << x <<" :"<< y;
return 0;
}
```

Output:

3:9
9:90

6. What would be the output produced by executing the following C++ code? **[5 Mark]**

```
int main() {
   int num = 212, i;
   for (i = 0; i < 4; i++) {
      cout << "212 >> " << i << " = " << (212 >> i) << endl;
   }
   for (i = 0; i < 4; i++) {
      cout << "212 << " << i << " = " << (212 << i) << endl;
   }
   return 0;
}
```

Output:

```
212 >> 0 = 212
212 >> 1 = 106
212 >> 2 = 53
212 >> 3 = 26
212 << 0 = 212
212 << 1 = 424
212 << 2 = 848
212 << 3 = 1696
```

7. What would be the output produced by executing the following C++ code? **[3 Mark]**

```
void my_fun(int *p, int *s) {
        s = p;
        *s = 23;
        int x = 5;
        s = &x;
}
int main() {
        int x = 3;
        int *p = &x;
        int *s;
        s = &x;
        my_fun(p, s);
    cout << "x = " << x << " *p=" << *p << " *s=" << *s;
        return 0;
}
```

Output:

```
x = 23 *p=23 *s=23
```

8. What would be the output produced by executing the following C++ code? **[5 Mark]**

```
void find(int a, int& b, int& c);
int main() {
        int one, two, three;
        one = 5;
        two = 10;
        three = 15;
        find(one, two, three);
        cout << one << ", " << two << ", " << three << " / ";
        find(three, two, one);
        cout << one << ", " << two << ", " << three << endl;
        return 0;
}
void find(int a, int& b, int& c) {
        int temp;
        c = a + 2 * b;
        temp = b;
```

Output:

```
5, 5, 25 / 35, 25, 25
```

```
        b = a;
        a = 2 * temp;
}
```

9. What would be the output produced by executing the following C++ code? **[5 Mark]**

```
void magic(int *sp, int *ep){
        while (sp >= ep){
                cout << *sp << " ";
                sp-=2;
        }
        sp += 5;
        cout << " / ";
        while (ep <= sp){
                cout << *ep << " ";
                ep+=2;
        }
        cout << endl;
}
int main(){
        int array[] = { 5, 4, 3, 2, 1, 2, 3, 0 , 4, 5 };
        magic(array + 9, array);
        return 0;
}
```

Output:

```
5 0 2 2 4  / 5 3 1
```

10. What would be the output produced by executing the following C++ code? **[5 Mark]**

```
struct Play
{
        int score, bonus;
};
void calculate(Play &P, int N = 10)
{
        P.score++;
        P.bonus += N;
}
int main()
{
        Play PL = {10, 15};
        calculate(PL, 5);
        cout << PL.score << ":" << PL.bonus << endl;
        calculate(PL);
        cout << PL.score << ":" << PL.bonus << endl;
        calculate(PL, 15);
        cout << PL.score << ":" << PL.bonus << endl;

        return 0;
}
```

```
11:20
12:30
13:45
```

Roll Number: _____ **Section: _____**

Q No. 3: Identify and describe the type of errors (syntax, logical or runtime) in the following code snippets and suggest how we could correct them. **[10 Mark]**

Code	Error description and Correction

`void input (int arr[][], int row, int col) {`
`  for (int i = 0; i < col; i++) {`
`  for (int j = 0; j < row; j++)`
`  cin >> arr[j][i];`
`  }`
`}`
 	
`int main() {`
`  int arr[3][2];`
`  input (arr, 3, 2);`
`  return 0;`
`}`
	[3 Mark]
	We must have to declare the second dimension in function header.

`void Sum(int a) {`
`  cout << a + 100 << endl;`
`}`
 	
`void Sum(int a; int b; int c = 10); {`
`  cout << a + b + c << endl;`
`}`
 	
`int main() {`
`  Sum('A');`
`  Sum('B', 30);`
`  Sum(20, 30, 90.5);`
`  return 0;`
`}`
	[3 Mark]
	Multiple errors:
	1. Semicolon after function.
	2. No definition exist for function with 2 parameter but in main we are calling the function with 2 parameters.
	3. Logical mistake; double values mapping to int variables.

`int main()`
`{`
`  int n,div=0;`
`  cout<<"Enter any number";`
`  cin>>n;`
`  div=n/0;`
`  cout<<div;`
`}`
	[2Mark]
	Runtime error division by zero.

`int count;`
`while (count < 100)`
`{`
`cout << count;`
`count++;`
`}`
	[2 Mark]
	Count not initialized.

Roll Number: _____ **Section: _____**

Q No. 4: …… **[10 Marks]**

Part I- Complete the code of the following program as mentioned in the comments and display the output of the highest and lowest values of the array after changing its values as given below.

10

```
#include<iostream>
using namespace std;

int main()
{
   const int SIZE=10;// Define correct variable for the size of the array.
   int numbers[SIZE] = { 3,4,7,9,8,11,6, 12, 14, 10 };

// The following code will change the values of the above array.

   for (int count = 0; count < SIZE; count++)
      numbers[count] = numbers[count] + 5 * 2;
// define variables to find the lowest and highest values.
int lowest = numbers[0];
int highest = numbers[0];
// write the code to find the highest value in the above array after changes.
for(int i=0; i<SIZE; i++)
{
if(highest <numbers[i])
         highest =numbers[i];
}
cout << "The highest value in the array is: " << highest << endl;
// write the output of the above statement here: ___24_____________
// write the code to find the lowest value in the above array after changes.
for(int i=0; i<SIZE; i++)
{
         if(lowest >numbers[i])
                  lowest =numbers[i];
}
cout << "The lowest value in the array is: " <<lowest;
// write the output of the above statement here: _____13_________
return 0;
}
```

Part II-Write the output of each statement of the following program as mentioned.

```
#include<iostream>
using namespace std;
int main()
{
   int i = 2;
   int array[i] = {5,2,3,10,7};

   cout << ++array[i] << endl; // write output here: _4_________

   cout << array[++i]++ << endl; // write output here: _10_______

   cout << array[i] << endl; // write output here: _____11_______

   cout << --array[i] << endl; // write output here: _______10____

   cout << array[i--] << endl; // write output here: _____10_____

   return 0;
}
```

ss

Q No. 5…….....… **[35 Marks]**

Part I - Pig Latin is a constructed language game where words in English are altered according to a simple set of rules.

1. If the word begins with consonant (non-vowel), Pig Latin takes the first alphabet, moves it to the end of the word and adds an ay at the end.
2. If a word begins with a vowel, it just adds "way" to the end.

For Example: "Banana" becomes "ananabay", "trash" becomes "ashtray", "eat" becomes "eatway", "omelete" becomes "omeleteway".

Write a program that reads a sentence as input and converts each word to "Pig Latin".

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Input in English:

One language is never enough.

Output in Pig Latin:

Oneway anguagelay isway evernay enoughway.

Solution:

```
#include<iostream>
using namespace std;

void Pig_latin(char input[],int s,int e)
{
	if (input[s] == 'a' || input[s] == 'e' || input[s] == 'o' || input[s] == 'i' || input[s] == 
'u'|| input[s] == 'A' || input[s] == 'E' || input[s] == 'O' || input[s] == 'I' || input[s] == 'U')
	{
		for (int i = s ; i < e; i++)
			cout << input[i];
		cout << "way ";
	}
	else
	{

		for (int i = s+1; i < e; i++)
			cout << input[i];
		cout << input[s] <<"ay ";
	}
}
int main()
{
    char arr[] = "One language is never enough";

    int start = 0, end = 0;
    int chk = -1;
    for (int k = 0; arr[k-1] != '\0'; k++)
    {
        if (arr[k]==' '|| arr[k] == '\0')
        {
            end = k;
                    Pig_latin(arr, start, end);
                    start = end + 1;

        }

    }
}
```

Part II - Write an algorithm to determine if a number n is happy.
A **happy number** is a number defined by the following process:

- Starting with any positive integer, replace the number by the sum of the squares of its digits.
- Repeat the process until the number equals 1 (where it will stay), or it **loops endlessly in a cycle** which does not include 1.
- Those numbers for which this process **ends in 1** are happy.

Return true *if* n *is a happy number, and* false *if not.*

Note that if a number is not a happy number, there will be an endless loop to this execution.

Goal: In this question, you are required to write C++ code that checks whether the number entered by the user is a happy number or not for 10 cycles/iterations only.

Input: n = 19

Output: true

Explanation:

$1^2 + 9^2 = 82$

$8^2 + 2^2 = 68$

$6^2 + 8^2 = 100$

$1^2 + 0^2 + 0^2 = 1$

Solution:

```
#include<iostream>
using namespace std;

int Happy_Number(int s)
{
	int a = s;
	int tmp;
	for (int i = 0; i < 10; i++)
	{
		int sum = 0;
		while (a != 0)
		{
			tmp = a % 10;
			sum = sum + (tmp * tmp);
			a = a / 10;
		}
		if (sum == 1)
		{
			cout << "Number is  Happy.\n";
			return 0;
		}
		else
			a = sum;
	}
	cout << "Number is Not Happy.\n";
	return 0;
}
```

```
int main()
{
	int n;
	int tmp;
	cout << "Enter a number :\t";
	cin >> n;

	if (n <= 0)
	{
		cout << "Enter positive number.\n";
		exit(0);
	}

	Happy_Number(n);
}
```